Financial Engineering Project

Making Concentrated Liquidity Pay

Risk-Sensitive Reinforcement Learning on Uniswap v3

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Abstract—Providing liquidity on Uniswap v3 looks passive, but it is not. An LP chooses a price range, earns fees while the price stays inside it, and loses whenever informed traders move the pool price against the quoted level. Those losses are not hypothetical: on many large pools the arbitrage losses borne by liquidity providers exceed the fees they earn. The LP is left with three decisions, all of them costly: when to recenter, how wide to keep the band, and how much of the position's exposure to hedge. We study those decisions as a risk-sensitive reinforcement learning problem, in a testbed built on two years of recorded on-chain trades. An exact replay engine provides the ground truth, a fast simulator provides the training throughput, and a parity harness keeps the two honest. The report asks four questions. Where does fee income fall short of the losses? Why does a standard learning algorithm fail on this task, and what repairs it? Can a single policy manage timing, width and hedging together and outperform the best rules written by hand on unseen data? And under which market environment does active management begin to pay? [PENDING: The final numbers and the last two answers will be filled in once the experiment runs complete.]

Index Terms—reinforcement learning, Uniswap v3, concentrated liquidity, market making, loss-versus-rebalancing, impermanent loss, risk-sensitive reinforcement learning.

I. Introduction

AUTOMATED market makers let anyone earn trading fees without quoting prices by hand. Uniswap v3 sharpened the offer: an LP chooses the price range it is willing to trade in, so a deposit of 100 000 USDC can earn the same fees as a much larger full-range position as long as the price stays inside the chosen band.[1] Narrowing the band raises the fee earned per dollar of capital, and that promise is what drew liquidity into v3.

The promise has a hidden price. Liquidity quoted at a price that has already moved is an offer that arbitrageurs will take. Every such trade pushes the pool toward the market price and leaves the LP holding more of the asset that is falling and less of the one that is rising. The loss does not reverse when the price comes back, because it is a transfer rather than a fluctuation. Milionis et al. gave this adverse selection cost its modern name, loss-versus-rebalancing (LVR), and showed that a full-range position loses value at a rate proportional to the square of volatility.[2] Fritsch and Canidio then measured realised arbitrage losses across large pools and found that they exceed the fees those pools earn.[3] Loesch et al. showed that concentration makes matters worse when the price leaves the band, since a narrow position can lose more than a plain full-range deposit.[4] Fee income, on its own, is not the LP's return.

An LP who wants more than a passive range has three decisions to make, and they interact. The first is when to recenter: acting early protects fee income, while waiting saves gas and slippage. The second is how wide to set the next band: a narrow band earns more per trade but leaves it sooner. The third is how much of the position's price exposure to hedge: a short position in the perpetual market removes directional swings, but it costs fees and funding. Each decision is an impulse, lumpy in cost and taken under uncertainty about volatility, order flow and regime. The objective is not simply to earn more, but to earn more per unit of risk taken. The decisions are lumpy in cost and made under uncertainty, which makes this a problem of impulse control rather than a static portfolio choice.[5]

Learning these decisions is a natural application of reinforcement learning, and recent work takes up parts of it. Zhang et al. train a hedged agent for v3 liquidity provision, but their hedge executes without cost or funding and the policy sees no regime information.[6] ZeroSwap learns to set a market maker's spread so that it tracks an external price, which is a different lever: it decides how wide the quoted price is, not where liquidity sits.[7] RAmmStein derives rebalancing thresholds that depend on the regime from a model of impulse control and reports large savings in rebalance count, but its net return excludes impermanent loss and it works with off-chain data for a single pair.[5] Chionas et al. study risk aversion in concentrated liquidity within a simulation and find that volatile, expensive markets push optimal bands wider and rebalances rarer, but they use no on-chain data and leave the hedge out.[8] None of these studies places the timing, width and hedge decisions in a single risk-sensitive policy, charges every friction on both the liquidity and hedge legs, and evaluates it on recorded on-chain trades against rules written by hand.

This report does that. We build a testbed on two years of recorded trades from two real ETH/USDC pools, with two engines: an exact replay backtester that computes fees in the protocol's own integer arithmetic, and a fast simulator that can be trained on for millions of steps. A parity harness compares the two on identical inputs, so the simulator can be trusted where speed is needed and checked where accuracy matters. The report then asks

four questions, in order.

- Where do fees fail? We decompose passive returns into fees, impermanent loss, gas and slippage, and locate the conditions under which the losses win.
- Why does naive learning fail? We document the collapse of a standard policy on this task, trace it to three interface problems, and measure what each repair buys.
- Can one policy do better? We train a risk-sensitive policy over timing, width and hedging and compare it out of sample with the best hand-written rules and a hedged benchmark.
- Where is active management worth it? We map the boundary between active and passive provision across volatility, fee tier, gas and capital.

The failures of the first policies are part of the result, not a footnote. A report that showed only the eventual policy would hide the most transferable lesson: in this domain, what the agent is allowed to see and what it is rewarded for matter more than the choice of learning algorithm.

The remainder of the report is organised as follows. Section II describes how a concentrated position works and why it loses. Section III places the study in the literature. Section IV states the decision problem. Section V describes the testbed and the failures that shaped it. Section VI fixes the protocol and the benchmarks. Section VII answers the four questions. Section VIII discusses what the evidence supports, and Section IX concludes.

II. How Concentrated Liquidity Works

A. A position as a bet on a range

Uniswap v3 indexes price by integer ticks. An LP picks a lower and an upper tick, and the deposit backs trades only while the pool price stays between them.[1] In protocol units the price is a raw square root, $s = \sqrt{1.0001^{\text{tick}}}$, which increases with the tick; the human price is $q(s) = 10^{d_1-d_0}/s^2$ USDC per WETH, so a lower tick means a higher ETH price. Inside a band $[s_a, s_b]$ the position holds

$$x(s) = \frac{L}{10^{d_0}}\left(\frac{1}{s} - \frac{1}{s_b}\right), \qquad y(s) = \frac{L}{10^{d_1}}\,(s - s_a)\,, \quad (1)$$

in token units, so its USDC value is

$$V(s) \;=\; x(s) + y(s)\,q(s) \;=\; \frac{L}{10^{d_0}}\left(\frac{2}{s} - \frac{1}{s_b} - \frac{s_a}{s^2}\right). \quad (2)$$

The formulas describe a simple behaviour. As the ETH price falls, the position sells USDC and accumulates WETH; as the price rises it does the opposite. At the edge of the band the position holds a single asset, and once the price leaves the band it stops trading altogether. The LP is always on the wrong side of the move, and a passive position cannot escape it.

Fees accrue in proportion to the position's share of active liquidity. Narrowing the band raises that share for a given deposit, which is why concentration is attractive, but it also shortens the expected time until the price reaches an edge. This tension between fee density and band lifetime sits underneath every decision in the report. Figure 1 shows the two effects together.

B. Two losses, one larger

The first loss is familiar. Impermanent loss (IL) is the shortfall of the position against simply holding the initial deposit. For a full-range position with equally valued legs, if the ETH price ratio is $r = q_t/q_0$, then

$$\mathrm{IL}(r) \;=\; \frac{2\sqrt{r}}{1+r} - 1, \quad (3)$$

which is second order in the price move; at $r = 1.2$ it is $-0.41\,\%$. On v3 the same expression applies only inside the band, and outside it the position is a one-sided holding, so IL depends on where the band was placed.[4]

The second loss is sharper. Compare the position with a portfolio that continuously rebalances to hold the same instantaneous exposure. A second-order expansion of the full-range value shows that the LP falls behind that portfolio at the rate

$$\mathrm{dLVR}_t \;=\; \frac{1}{8}\,\sigma^2\,V_t\,\mathrm{d}t, \quad (4)$$

which is the loss-versus-rebalancing of Milionis et al.[2] Adverse selection therefore grows with the square of volatility and with position size, and fee income must exceed it for passive provision to be viable. Concentration raises fee density but not the fee rate, so it helps only when the extra fee share outweighs the extra exposure to arbitrage. The measurements in Section III find that, on many large pools, it does not. Figure 2 shows the shape of both losses.

C. The decisions and the risk

An LP that wants to improve on the passive outcome must decide when to act, where and how wide to place the next band, and how much market risk to carry. The decisions interfere. A wider band delays the next costly recentring but earns less per trade; a hedge removes directional variance but pays funding and perp fees; acting sooner protects fee income but multiplies transaction costs. Because acting is lumpy, the problem is one of impulse control,[5] and because the outcome is uncertain, the right measure is not profit but profit per unit of risk. We use two such measures throughout: the annualised Sharpe ratio and the conditional value-at-risk (CVaR) at the 5 % level, a coherent risk measure that can be optimised through its convex representation.[9] Figure 3 makes the sequence concrete.

III. Related Work

The economics of passive provision is now well understood. Milionis et al. formalised the loss a constant-product LP suffers against a delta-rebalanced portfolio and derived the rate $\frac{1}{8}\sigma^2 V$ for a full-range position; the loss appears on any price path, informed traders or not,

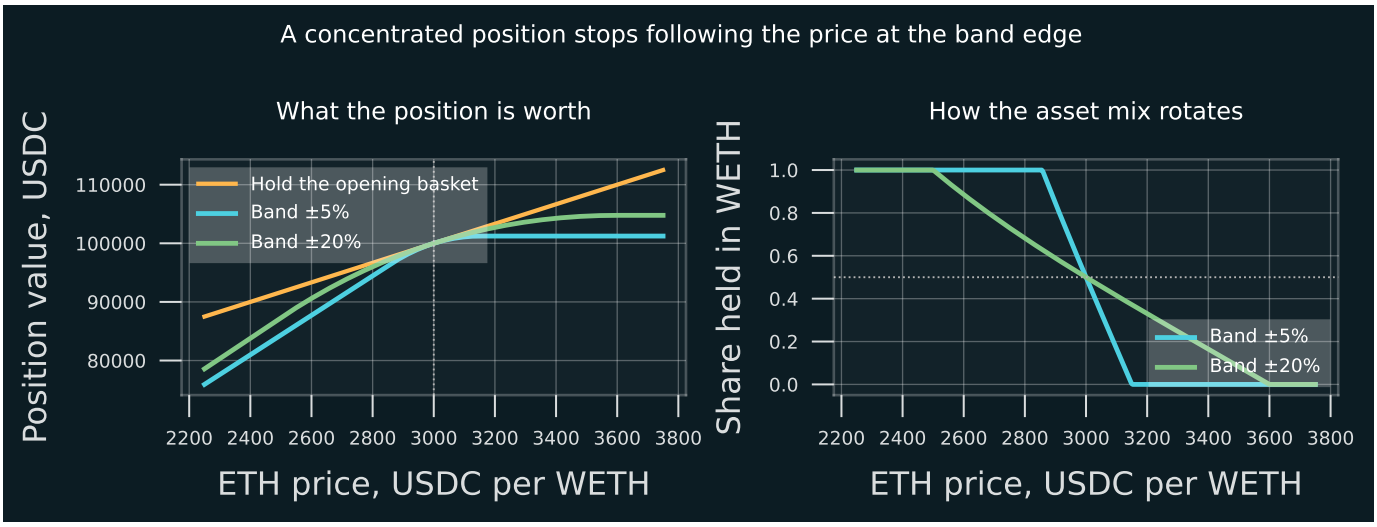


Fig. 1. A concentrated position versus a passive holding of the same opening basket. Left: the position tracks the basket while the price sits inside the band and stops following it at the edges, so it gives up the upside of a rally while keeping the downside of a fall. Right: the asset mix rotates from WETH to USDC as the price rises and freezes at the edge. Illustrative parameters: 100 000 USDC entered at 3000 USDC per WETH.

and protocol fees and the choice of curve shift only the balance.[2] Fritsch and Canidio measured the losses on real pools and found that arbitrage losses typically dominate the fees earned, with passive full-range positions outperforming concentrated ones in their sample.[3] Loesch et al. studied v3 positions directly and showed that concentration magnifies impermanent loss when the price leaves the band.[4] Together these results define the terrain: gross fee yield is not an LP's return.

A second line of work learns the LP's decisions. Zhang et al. train a hedged double deep Q-network and report large gains over baselines, but the hedge is assumed frictionless and the policy sees no regime features.[6] ZeroSwap learns the spread of a market maker so that quoted prices track an external reference, an orthogonal lever to range placement.[7] RAmmStein poses regime-dependent rebalancing thresholds as a problem of impulse control and reports that rebalancing lazily when the regime warrants it improves net ROI with far fewer rebalances, though its net metric excludes impermanent loss and the study relies on off-chain data for one pair.[5] Chionas et al. treat concentrated liquidity as a risk-aversion problem in simulation and find that higher volatility and gas widen optimal positions and reduce rebalancing, while a risk penalty trims the adverse tail; their analysis uses no on-chain data and no hedge.[8] Classical market making supplies the risk-sensitive vocabulary we use, but its state is a limit order book and its actions are quotes, not range geometry.[10]

The learning machinery is standard. We use proximal policy optimisation with clipping,[11] implemented through Stable-Baselines3 on Gymnasium environments,[12], [13] and evaluate risk with CVaR at the 5 % level through its Rockafellar–Uryasev representation.[9]

What is missing is one study that trains a single policy over the joint timing, width and hedge decisions, charges every friction on both legs, and evaluates it on validated on-chain replay under a risk-sensitive metric. That is the gap this report fills.

IV. The Liquidity Provider's Decision Problem

We model the LP's problem as a risk-sensitive Markov decision process with impulse control. Decisions are taken every ten minutes over an episode of thirty days. At each decision the agent sees a fixed observation, chooses between holding and acting, and receives a reward that measures how far the strategy's value has moved beyond a passive benchmark, net of every cost. The process is partially observed: the volatility regime and order flow are latent, and the agent sees only the history summarised in the observation.

A. State

The observation contains 24 features, all computed from information available at the decision time. The market group covers the current price, recent returns and realised volatility, the pool's active liquidity and fee tier, the gas price and the regime label. The position group covers the value of the live position, the fees it has accrued, its loss against the initial basket, and the distances from the current tick to each edge of the band together with the band width. The hedge group covers the short size relative to the position's ETH exposure, the equity in the hedge account, and the most recent funding rate. The band geometry deserves emphasis. The action space is written in tick spacings relative to the current tick, so a policy that cannot see how close the price is to an edge cannot time a rebalance at all. Heavy-tailed features such as price,

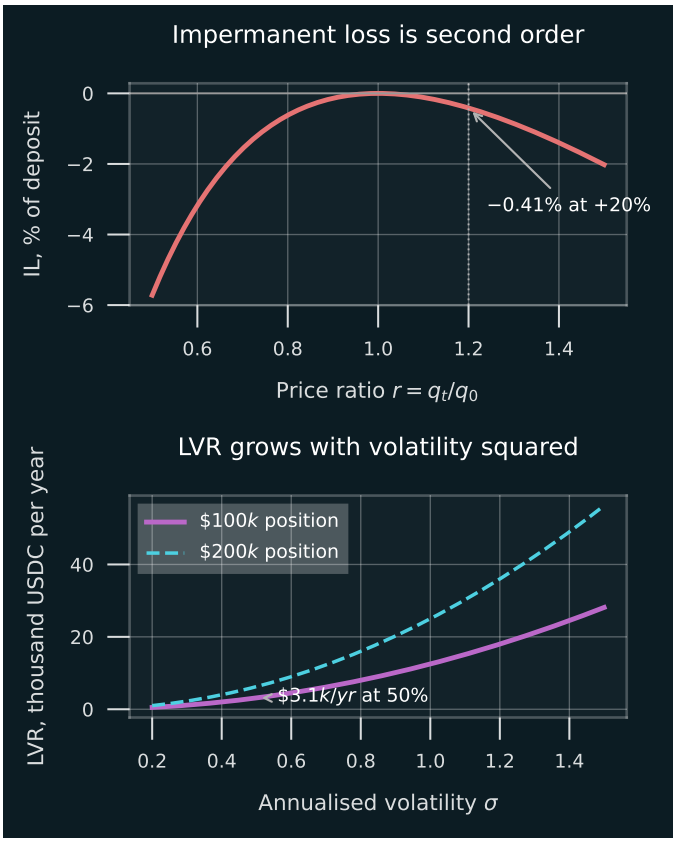


Fig. 2. The two losses. Top: impermanent loss against the price move, second order and small for a full-range position. Bottom: the LVR rate against volatility for a 100 000 USDC and a 200 000 USDC position, showing both the quadratic dependence on volatility and the linear dependence on size.

liquidity and gas are compressed with a signed logarithm so that every dimension reaches the network on the same scale.

B. Action

At each decision the agent selects a factored action

$$a_t = (\kappa_t,\, c_t,\, w_t,\, h_t), \tag{5}$$

where κ_t decides between holding and rebalancing, $c_t \in \{-4, \dots, 4\}$ shifts the centre of the new band in tick spacings around the current tick, $w_t \in \{1, 2, 5, 10, 25, 50, 100, 250, 500, 1000\}$ is the band half-width in tick spacings, and $h_t \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is the fraction of the LP's ETH delta carried by the short hedge, so that $h_t = 1$ matches the position's beta. The liquidity head alone has $1 + 9 \times 10 = 91$ options. In the experiments of this report the hedge ratio is held at a configured value while the liquidity head is learned; promoting h_t to a learned head is the joint-policy extension evaluated in the last two questions.

C. Reward

A rebalance converts inventory and pays for the privilege. Writing V_t for the position value, H_t for the fixed value of the opening deposit held passively, and the hedge cash flows as mark PnL π_t, funding ϕ_t and perp cost $\tau_t \le 0$, the reward is

$$r_t = \frac{1}{W_0}\Big(\text{fees}_t - \text{gas}_t - \text{slip}_t + \Delta(V_t - H_t) + \pi_t + \phi_t + \tau_t - \lambda\,\hat{\sigma}_t\Big), \tag{6}$$

where fees are non-negative income, gas and slippage enter as positive cost magnitudes, W_0 is the initial capital, $\hat{\sigma}_t$ estimates recent reward volatility, and $\lambda \ge 0$ dials risk aversion ($\lambda = 0$ by default). The excess is measured against the fixed opening basket: H_t is never rebased at a rebalance, so the episode sum equals the excess over holding that basket, which is the quantity the evaluation reports. Dividing by W_0 keeps value targets at a workable scale without hiding any cost.

D. Risk and benchmarks

The learning objective is the discounted return adjusted for tail risk. With losses $L = -G$ defined on the episode return G, the conditional value-at-risk at level α is

$$\text{CVaR}_\alpha(L) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{1-\alpha}\mathbb{E}\big[(L-\tau)^+\big] \right\}, \tag{7}$$

and the policy maximises $\mathbb{E}[G] - \lambda\,\text{CVaR}_\alpha(L)$ with $\alpha = 0.95$.[9] Evaluation reports the annualised Sharpe ratio and CVaR at 5 % as primary statistics, with raw PnL and excess over HODL as secondary ones. Comparisons use paired per-window differences and confidence intervals across seeds.

The hedge is a short perpetual position sized as a fraction of the LP's delta. Its mark PnL over a step is the negative product of the short notional and the price change, plus funding and the taker cost of adjusting the position. Three reference points make the hedge leg interpretable: the unhedged LP, the deterministic overlay, and a hedged HODL portfolio with matched beta. The last is the benchmark that prevents us from declaring success against a weaker opponent, a policy that merely removes directional risk must still beat a passive portfolio with the same risk profile.

V. Building a Trustworthy Testbed

A study of this question needs a world in which the LP's costs can be measured rather than assumed. That world has to meet two demands at once: it must be exact enough that a reported fee or loss is the real one, and fast enough that a policy can make millions of decisions. No single engine satisfies both, so the testbed uses two and checks them against each other. Figure 4 shows the flow.

A. The record

The data are the raw logs of two real ETH/USDC pools: the 0.05 % pool used as the primary market, and the 0.30 % pool used as a fee-tier contrast. The pipeline turns those logs into one ordered event tape of swaps, mints, burns and collects, and joins it with three auxiliary feeds: one-minute

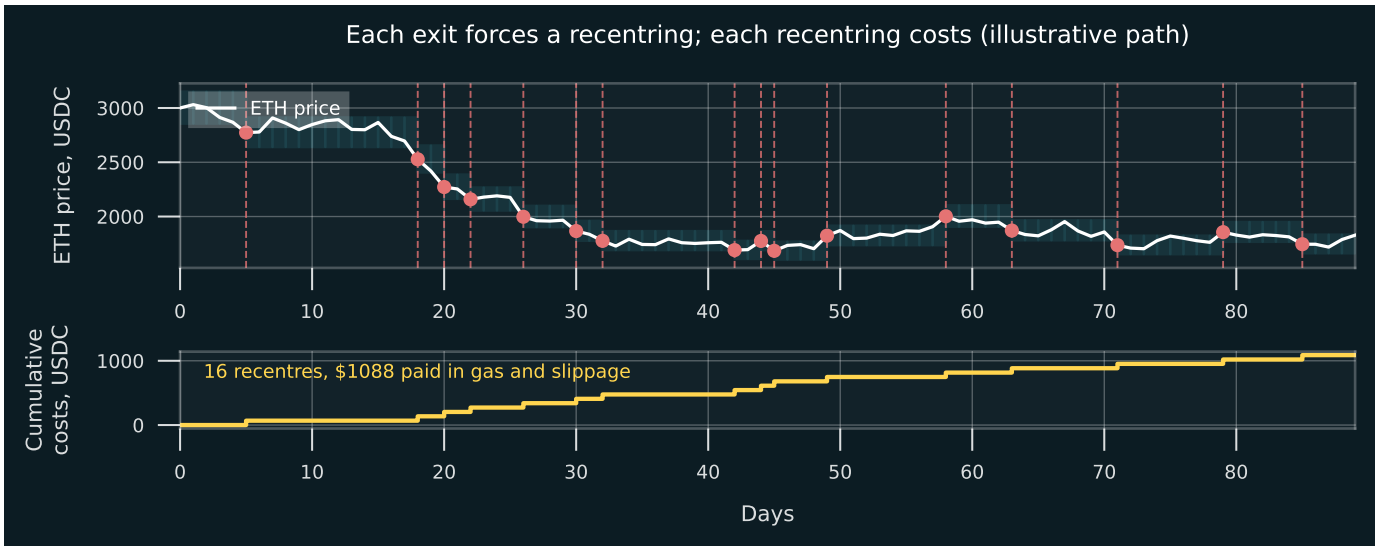


Fig. 3. Why acting is a decision rather than a rule. An illustrative price path with a ±5% band: each exit forces a recentring, and each recentring adds gas and slippage to a cost that only ever grows. The path is illustrative and the costs are simplified; the study charges per-block gas and slippage on the inventory actually swapped.

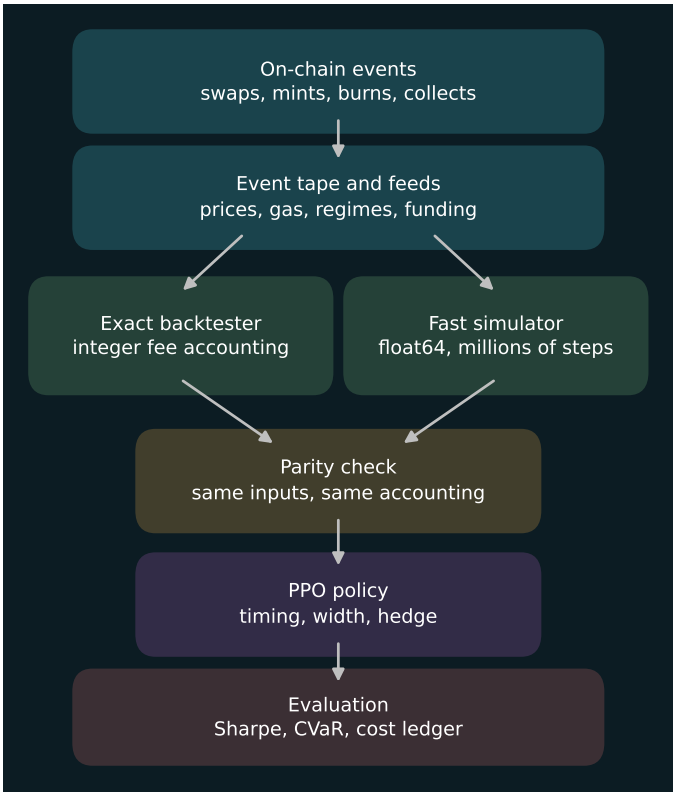


Fig. 4. The testbed. Recorded events feed one tape; two engines consume it, an exact backtester for ground truth and a fast simulator for training; the parity check reconciles them before a policy is trusted.

reference prices, per-block gas prices, and regime labels computed from thresholds fixed in advance. A fourth feed, the funding rate of a perpetual margined in USDT, prices the hedge. The window runs from January 2022 to December 2024. Training uses 2022 and 2023; 2024 is held out. The primary pool's training tape alone holds about 4.86 million events, which is why speed matters so much later. Every read goes through one interface that enforces block ordering and the train/eval boundary, so an observation at time t cannot accidentally see a bar from $t+1$.

B. The exact engine

The backtester replays a frozen policy against the tape in the protocol's own integer arithmetic. Fees come from the pool's recorded fee-growth accumulators, exactly as the contract computes them; liquidity events update an external tick lattice seeded from a boundary snapshot; swaps are apportioned across the ticks they cross. The ledger therefore satisfies an accounting identity by construction: fees plus impermanent loss, minus gas and slippage, equals the strategy's excess over holding the opening deposit. A backtested policy is frozen by construction because the interface exposes no way to update it, and every ledger records the configuration and seed that produced it.

C. Two engines, one check

Training on the exact engine would be far too slow, so the simulator mirrors the same economics in floating point. Its agent is marginal: the position never moves the price, and its fee share is the fraction of active liquidity it supplies. The simulator can run on the recorded events or on a calibrated regime-switching jump process fitted to the training years, and it executes each action at the price observed before the interval advances, as a live LP would.

Two numeric worlds are only useful if their disagreement is bounded. The parity harness feeds both engines the

same prices, positions and gas series, and then compares fees, impermanent loss and PnL under a stated tolerance. Since the simulator's fee engine was made exact, the remaining drift on real data is tiny: mean fee differences on the order of 10^{-7} of notional per seed, and impermanent loss matching exactly. The harness originally exercised only the passive policy, which never rebalances; it now also runs a policy that does, which is the path the fee and loss results depend on. A separate suite of look-ahead probes attacks the composed system at the train/eval seam.

D. The failures that shaped the design

The first two training runs produced policies that did nothing useful. Every seed collapsed to one deterministic action, some always holding and some always repeating a single rebalance, and the measured advantage barely distinguished the actions. The collapse was not a tuning problem. The observation contained no information about the band, so a policy could not tell how close the price was to an edge or how far a recentring would have to move. The reward measured the loss against a rolling entry basket, so after a rebalance it no longer matched the metric the evaluation reported. And every training episode replayed the same window of thirty days, so the agent never saw a different market. Two cost defects surfaced while fixing those: slippage was charged on the whole position value instead of the inventory actually swapped, and the marginal fee credit was not capped when the agent supplied only a small share of liquidity.

Each repair is small. Together they were the difference between a policy that can learn and one that cannot, which is why the failure itself is treated as a result rather than an implementation note.

VI. Experimental Design

The protocol is fixed before any result is inspected. The study uses the 0.05 % primary pool and the 0.30 % contrast pool described above, with training on 2022–2023 and evaluation on 2024. Training episodes are windows of thirty days sampled from the training years; evaluation tiles 2024 into non-overlapping windows of thirty days, and no episode crosses the split boundary. Each episode runs at a cadence of ten minutes with 100 000 USDC of capital, which keeps the agent small relative to pool liquidity. Every configuration is trained with five seeds and one million timesteps per seed.

A. Benchmarks

A learned policy that beats only the weakest rule written by hand has demonstrated nothing. The comparison therefore includes the full ladder in Table I, a best static width chosen on the training split and then frozen, and a hedged HODL portfolio with matched beta. The learned policy must beat the strongest of these, not merely a passive holding.

TABLE I
Benchmark policies. The learned policy is compared with all of them; a positive claim requires beating the best static width and the hedged HODL with matched beta, not just hodl.

Benchmark	Rule
hodl	Hold the initial token mix; never provide liquidity.
passive_narrow	Deploy a ±5 % band once; never recenter.
passive_wide	Deploy a ±20 % band once; never recenter.
full_range_v2	The Uniswap-v2-equivalent full range.
tau_reset	Recenter whenever the price exits the live band.
cost_aware	Recenter at most daily, and only when the estimated fee gain exceeds gas.
Best static width	The ladder width with the best Sharpe on the training split, frozen for test.
Hedged HODL	Initial basket plus a short perp matched to its beta, with funding and perp costs.
Learned policy	PPO over the joint (timing, width, hedge ratio) action space.

B. Metrics

The two primary statistics are the annualised Sharpe ratio, computed on each episode's equity curve and paired across seeds and windows, and CVaR at the 5 % level, the average of the worst 5 % of episode returns pooled across seeds. Both are reported with bootstrap confidence intervals; because the tail is thin at thirteen windows and five seeds, the worst window is reported alongside them. Secondary statistics are raw PnL, excess over HODL, annualised return, Sortino ratio, maximum drawdown, rebalance count, and the cost ledger of fee income, gas, slippage and realised impermanent loss. Per-regime tables stratify by each window's dominant label: bull, bear, sideways or high volatility.

C. Ablations and the boundary

The failure modes are ablated one at a time: band geometry removed from the observation, training windows held fixed instead of resampled, the hedge disabled, the risk penalty toggled, and the regime features removed. The cost decomposition is ablated over frictions (full model, no gas, static fee, no impermanent loss) and repeated on the 0.30 % pool to test whether a higher fee tier buys a longer viable range. For the boundary question the runner sweeps realised volatility, volume relative to liquidity, fee tier, gas price and capital, evaluating each point on replayed windows rather than a fitted process.

The decision rules are stated in advance. The learned policy earns a positive verdict only if its mean Sharpe beats the best static width and the hedged HODL with a paired 95 % confidence interval excluding zero, and its CVaR is no worse than the best heuristic's. Where the evidence is negative, the negative result is the finding.

VII. Results

[PENDING: This section is deliberately a stub while the multi-seed experiments run. Each subsection states the question and the shape of the evidence that will answer it; no number is written until the locked runs exist.]

A. Where do fees fail?

The first question is economic. We decompose the passive benchmarks into fee income, impermanent loss, gas and slippage, and locate the conditions under which the losses exceed the fees. [PENDING: Report the decomposition on both fee tiers, the cost of greedy against lazy rebalancing, and where the sign of excess over a passive holding flips across regimes.]

B. Why does naive learning fail?

The second question is methodological. A standard policy, trained on the original interface, collapsed to a single action on every seed. We trace that collapse to what the agent could see, what it was rewarded for, and the windows it trained on, and we measure what each repair contributes. [PENDING: Report the pre-fix collapse and the per-fix ablations, with the evidence for each diagnosis.]

C. Can one policy do better?

The third question is the central one. [PENDING: Report out-of-sample Sharpe and CVaR for the learned policy against every benchmark, paired per window and across seeds, together with the action distribution and the decision boundaries it recovers.]

D. Where is active management worth it?

The fourth question turns the result into a map. [PENDING: Report the boundary along which active and passive provision achieve equal performance under risk, as a function of volatility, volume relative to liquidity, fee tier, gas and capital, with the analytical LVR threshold overlaid.]

VIII. Discussion

[PENDING: The interpretive discussion is deferred until the locked runs land: it will connect the four answers to the active/passive boundary and to the design lessons for learning in this setting. The validity concerns below do not depend on the results.]

The simulator replays recorded events, so prices, liquidity and fee growth are real rather than modelled. Its marginal-agent assumption is the main simplification: the position never moves the price, and its fee share is proportional to active liquidity. That is the standard replay assumption and it is recorded with every run, but it cannot capture queueing at the tick level or the impact of an unusually large position. The gas model charges representative per-action units rather than per-transaction receipts, so gas is a defensible floor rather than an exact quote, and the hedge uses a perpetual proxy whose funding and basis are representative rather than exact for a specific venue.

Several threats bound what the study can claim. The evidence covers one asset pair at two fee tiers, so the boundary is measured within that scope and does not automatically transfer to other pairs. CVaR at 5 % averages a handful of episodes, so it is reported with bootstrap intervals and the worst window rather than treated as decisive on its own. The regime labels come from thresholds fixed in advance; they organise the reporting and anchor the training process, but they are a labelling device, not a discovered state. The boundary sweep compares several axes at once, so paired intervals on a pre-registered primary metric carry the inference, and any secondary win is labelled as such.

The parity check bounds the distance between the simulator and the backtester, not the distance between the backtester and reality. Where the data allow, the external part of the gap is quantified by reconciling fees and collects; where they do not, latency, order-flow privacy and MEV are discussed rather than measured. We therefore claim results about performance under risk on recorded market states under a stated cost model. We do not claim deployable profitability, optimality of the action grid, or that the policy would survive market conditions absent from 2022 to 2024.

IX. Conclusion

The story began with a simple bargain: concentrate liquidity in a price range, earn more fees while the price stays inside it. The bargain hides a cost. The same position that earns the fees is the position that loses to informed order flow, and on real pools that loss can exceed the income. An LP who wants to do better than accept it must decide when to recenter, how wide to stay and how much risk to hedge, and each decision is expensive and uncertain.

This report set out to measure whether a risk-sensitive learned policy can make those decisions jointly better than the best rules written by hand, and to find where active management stops paying. Answering that required a world in which costs are measured rather than assumed, so we built a testbed from two years of recorded trades, with an exact replay engine for ground truth and a classical simulator for training, reconciled by a parity harness. The study then asked four questions: where fees fail, why a standard agent fails, whether a learned policy wins out of sample, and where the boundary between active and passive provision lies.

[PENDING: This paragraph will close with the measured verdicts once the experiments complete. Its shape is fixed: what the four answers were, which of them held, and what the boundary looked like.]

The boundary of the study is also a map for future work. The action space can be extended with continuous width and an explicit exit decision. The boundary sweep

can cover more pairs and fee tiers. Latency, private order flow and the exact funding of a real venue can be modelled rather than proxied. Each extension is a re-run of the same testbed rather than a redesign of it, which is the point of building the testbed before asking the question.

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